

MTH209: Assignment 7

Debugging assignment

Consider the Central Limit Theorem for a Gamma distribution. For $X_1, X_2, \dots, X_n \sim \text{Gamma}(\alpha, 1)$, we know that $E[X] = \alpha$. We want to find the distribution of the test statistic t where:

$$t = \sqrt{n}(\bar{X}_n - \alpha)$$

Our goal is to write a function `MY.CLT_simulation` that takes a shape parameter `alpha` and returns a vector of simulated statistics for 100 repetitions with $n = 50$. Below is such a function, but it has errors:

```
MY.CLT_simulation <- function(alpha)
{
  reps <- 100
  n <- 50

  store <- numeric(length = n)

  for(r in 1 to reps)
  {
    sample <- rgamma(n, shape = alpha)
    store <- sqrt(n)*(mean(sample) - alpha)
  }

  return(store)
}
```

Your job in this assignment is to fix ALL the mistakes and make the function return a valid numerical vector of length 100.

INSTRUCTIONS: copy and paste **ONLY** the function `MY.CLT_simulation` in your `assignment7.R` file in the GitHub repository. **NO PARTIAL MARKING** for this assignment.